ALPHA 1 → ALPHA 2 → ALPHA 3 → THE GENOME · WRITTEN 2026-09-19 · EVERY NUMBER READ FROM THE FILES THAT DAY

The Alpha Research Atlas

Every research input, synthesis, experiment and mechanism since Alpha 1, joined by what each rests on and what each proved or broke; where the organisms fail and why; the theory behind each failure and its fix; and the plan that follows. Built on Ahmad's word of 2026-09-19 14:52.

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0 · The state the atlas was written from

RESEARCH INPUTS FROM AHMAD

≈ 34 2026-08-13 → 09-19: the 21 pastes the 09-06 deep research inventoried, and ≈ 13 items since (notes, diagrams, PDFs, tweets, the AGI Q&A)

SYNTHESES, PREREGS & REPORTS WRITTEN FROM THEM

≈ 150 Alpha 1: 14 proposals + 8 research files · Alpha 2: 10 proposals + 67 preregs + 26 reports · Alpha 3: R1–R6, 5 reviews, 2 research files, 5 digests, 8 designs, the gap map

ALPHA 3 EXPERIMENTS (DIRECTORIES)

56 PASS 13 · FAIL 32 · VOID 3 · INCONCLUSIVE 3 · NOT RUN 5; 22 more queued with no directory

ALPHA 1 · ALPHA 2 MECHANISMS JUDGED

27 · 36 worked / partial / failed, each with its evidence file

Alpha 1 (08-17 → 09-12) was a self-improving trading brain: a frozen judge/generator split over a bounded genome. Alpha 2 (08-29 → 09-12) was a Python-scheduled, C++-bodied organism judged at every step by a frozen judiciary; on 09-07 it was rebuilt in one day as a whole creature (six laws) and folded back. Alpha 3 (09-12 → 09-19) is the creature carried forward under a byte-identity gate, with organs added at its edge. All three are closed today: Alpha 1 and 2 retired 09-12 on Ahmad's word; Alpha 3 stopped 09-19 14:44 on his word after a crash loop and two days on one question. Every claim below names the file it was read from; the six read-only sweeps that produced the catalogue are kept verbatim in `docs/atlas-sources/`.

1 · The connection map

Time runs left to right. His inputs are the top lane; each organism is a lane; the genome is the lane that starts today. Solid arrows carry a mechanism forward; red arrows carry a failure that recurred. Colour is the verdict, not the era.

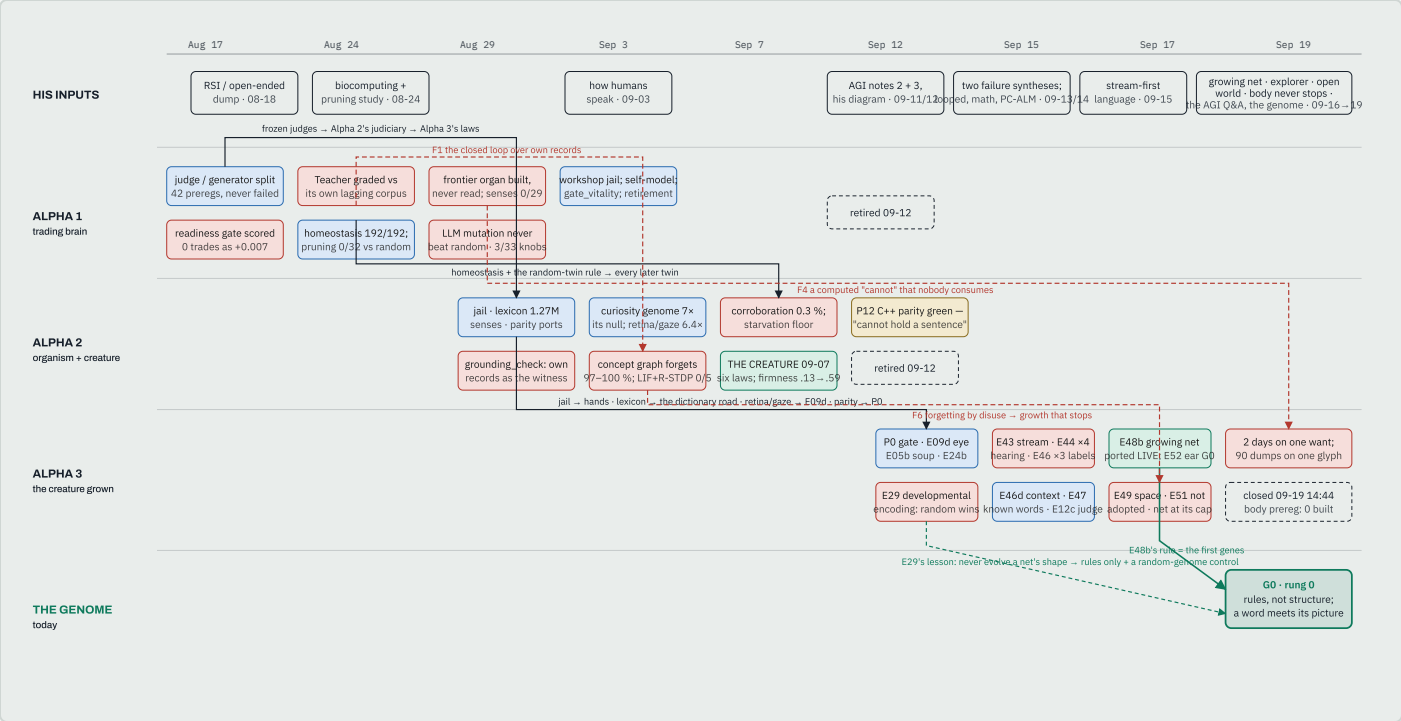


Figure 0. The lineage. Blue = worked, red = failed, amber = green gates that proved nothing, green = the living thread. The three red arrows are the failures that crossed organisms without changing shape; §3 names all ten.

worked / PASS failed green gates, no capability alive, carried forward

1.1 · His inputs (the sources everything rests on)

DATE	WHAT HE SENT	WHAT IT BECAME
08-13 → 08-29	five research pastes (mutation spectrum, an HH+STDP C++ sketch, the Alpha 2 decision, an eight-part chat research); the RSI/open-endedness dump (08-18); the biocomputing summary and his own synaptic-pruning study (08-24); the neuron-function and other-organisms directives	Alpha 1's plasticity program (M1–M6), the frontier organ, the belief organ; Alpha 2's founding program; the 09-06 deep research §1 inventory (21 pastes)
09-02 → 09-03	the T1–T13 research paste; «how humans speak»	P11 THE GROWTH PROGRAM (neural ladder, curriculum G0–G4); speech_v2's road; GROWTH-PROGRAM Addendum 3
09-06 00:50	the deep-research order (brain, animals, verify live, experiment, plan)	ALPHA-2-DEEP-RESEARCH (90 KB): Physarum 5/5 exact, run-and-tumble, the concept graph forgets 97–100 %,

		shelf 0.29 s/book; the four waves; six decisions
09-11 → 09-12	AGI notes part 2 (Quran 96:1–5, 2:31, 17:85; §A–I), part 3 + his diagram (a loop upstream of language), the fly tweets, Looped Flows	R1–R6 (six research files, 42 tests, the E-series queue), ALPHA-3-PLAN, the five digests of Alpha 1/2
09-13 → 09-14	two failure-research syntheses (eight joins; eleven threads); looped reasoning + development (two images); candidate mathematical frameworks (PDF); PC-ALM (PDF); FlashREINFORCE	four REVIEWs: E18's unblock VOID, E24b PASS, E16c FAIL; Decisions 7/8; E27–E35 queued
09-15 17:58	stream-first language (Saffran; the superior colliculus; McGurk)	the direction corrected: words come from the stream → E43/E43b/E44/E46 series; RESEARCH 09-16 on the failing organs
09-16 → 09-17	«the growing and shrinking neural weights»; «let him be an explorer»; «the own record is caging a bird»; «eyes, ears, thinking never stop»; his 3-D-space idea	E48/E48b (the growing net, live), D30, BLUEPRINT ADDENDUM 1 (the open world), FUSED-SENSES-PREREG (the body), E49
09-19	the AGI Q&A (8 sections, 9 open theories); «close alpha 3»; «a small genome experiment in python»; «gather all the research»	the GAP MAP; this atlas; G0

1.2 · Alpha 3's experiments — every directory, with what each rests on and feeds

Fifty-six directories. Read the verdict column with the reason beside it: a FAIL here is a measured fact with a named cause, and thirteen of them named the mechanism the next experiment fixed.

ID	QUESTION	VERDICT	THE NUMBER	RESTS ON → FEEDS
E01	independent corroboration on the shelf	PASS	1.28 % of 34,730 claims (twin 0.042 %); later 0/214 truly independent	→ E18-small, the corroboration bar
E02	certificate or silence: 0 false accepts?	FAIL	15/40 false accepts; the gate byte-reproducible	R1 → E11, E18, E27
E03	tropical trait ladder; VSA belief algebra	FAIL	ladder −37.4 pp; AUC 0.687 vs 0.75	R3 → E09a's gate, E31
E05 → E05b	the BFF soup: self-replicators with no fitness	VOID → PASS	0 of 20 seeds grew replicators on three port bugs (E05/VERDICT.md) → E05b 8/20 transitioned; RAND 0/20	R4 → E10 (open-endedness)
E06 · E06b	are his dispositions repeatable character?	INCONCL. · FAIL	the most repeatable reading is a scheduler alternation	R2 → journal the eligible set

E07a	Hopfield / HDC / NG-RC	FAIL (HDC-B pass)	NG-RC diverges on the integrator, not α; HDC publisher recovery R 1.0	R3 → E17, E28, E31
E08	a grown sentence realiser	FAIL	grammaticality 11.6 % vs 90; nonsense the checker passed	→ the meaning judge E12
E09a/b/c/d	the eye: slowness; EWC vs rehearsal; the tower; the readout	F P F PASS	E09d: 84.5 / 80.8 at 7,725 params = the eye organ	R2 → E25, E29, E39, E46, E51
E11	revise MEANING before rendering?	FAIL conj. / PASS the claim	upstream beats downstream by 47.5 pp	his diagram → E27, E30
E12 · 12b · 12c	a learned meaning judge	F F PASS	gates_only 7 clauses to 6 across three disjoint draws	→ E16, E36 (the live judge)
E14 · E21	the desk: a shell from pixels; self- vs world-caused change	PASS PASS	18/18; AUC 1.0 at 57 % noise	→ E28, E42 (his own computer)
E15 · E22	mushroom-body sparse expansion; its unrun arms	FAIL · PASS	dense beats sparse; verification-GATED plasticity necessary (+25.9 pp)	R5 → the tower is dense
E16 · 16c ×3	can any embedding beat lexical overlap?	FAIL ×4	from-scratch −10.9 pp vs Jaccard; MiniLM first to beat overlap (BA 0.941)	→ D11, D13, E36 (MiniLM sealed)
E17 · 17b	random-feature readouts	F · PASS	return collapses after 16×; invariance from variety, not size	closes the line
E18-small	an independent second witness?	INCONCL.	0 of 214 claims	his syntheses → D12(b)
E20	Tsetlin machine as explainer	FAIL parity	clause stability 0.738 vs 0.051	his syntheses → E20b
E24 · 24b	equation discovery on GoL, Lenia, klines	INCONCL. · PASS	GoL exact 512/512; stability selection finds the plant 6/6, the 1-SE rule 0/6	his synthesis A → the estate's selection rule
E25	keeping without an oracle	PASS	retention scales with rehearsal ROWS	→ E46/E48 buffers
E27 · E28	a halting comparator; a stateful world organ	FAIL FAIL	depth buys cost, not accuracy; 97 params match 8,000	his looped-reasoning paste → E30; E21
E29	**a developmental encoding grows the eye tower**	FAIL	grown 12.4 pp behind direct; **random θ beats evolved θ**	his development paste → the precedent G0 is built against

E36	seal the MiniLM judge	F → sealed at option A	FA 5.19 % vs 5; recall 60 %	D13 → the live meaning organ
E43 · 43b · **E47**	the stream cuts itself	F F PASS	boundaries 2.08× twin but the wrong grain; known words + statistics 12.6× twin	his stream-first study → E52; not wired
E44 ×4 → **E52**	hearing tied to the word	VOID ×2 F ×2 → G0 PASS	plants 0.91/0.88/0.93/0.84 under a 0.99 bar → 1.000 on 553 held-out; the gate exercises loudness only	→ G1 (CMUdict); the ear gene
E46 · 46b · 46c → **E46d**	the eye learns a second script without losing the first	F ×3 → PASS	the 18 shared glyphs were a LABEL CONFLICT; context cures it (g2 adopted)	his growth order → D31 → E48
E48 · **E48b**	the growing / shrinking net	F · F (digits) — rule LIVE	decay ×0.064 and a least-used cap forgot Latin; budget + no decay retains; byte-exact port	his 18:53 word → the live organ → G0's genes
E49 · E51	units in 3-D space; richer eye channels	FAIL FAIL	no sparsity, no modules, 2.3× cost; channels carry (1.000 vs 0.500) at 5.1× cost	his 3-D idea; his eye-traits word → a cheaper carrier
E50 · B2 · D31 · D33	the web's regions; the index pin; the family answer set; a made name to the world	NOT RUN / live unscored	D31: digits .69→.79, handwritten .19→.66	—

1.3 · Alpha 1 and Alpha 2 — the mechanisms, judged

ORGANISM	WORKED (WITH THE EVIDENCE)	FAILED (WITH THE CAUSE)
Alpha 1 27 JUDGED	judge/generator split (42 preregs, never failed a control) · pre-registration with appended amendments · cluster-SE fix (FP 72.8 % → 5.7 %) · alias/support control (real vs shuffled 51×) · homeostasis M3 (192/192 runaways contained) · the random-twin rule on pruning · the frontier organ's exhaustion typing · a sandboxed own-build workshop (9/9 hostile drafts refused) · self-model (reflection only) · gate_vitality (a gate that can never pass) · taught-vs-verified curriculum · immune forward-attribution audit · the retirement pattern (seal, delete nothing)	the entry rules self-contradicted → 0 entries in 117,439 · the readiness gate scored 0 trades as +0.007 (won by abstaining) · the Teacher graded against a corpus 24 h behind its advice · mutation opened 3 of 33 knobs · the LLM mutation operator never beat random (16,371 trials, 0 promotions) · wants→senses unlocked 0 of 29 · the frontier organ never read by the proposer · the discovery pipeline's own FDR 83–89 % · usage-pruning 0/32 vs random · plasticity 0/63 arms (the ceiling was the FEATURE VECTOR)
Alpha 2 36 JUDGED	the jail (Landlock + seccomp; Alpha 3's hands) · the parity/oracle port discipline (0 mismatches over 82 files) · lexicon (1.27 M senses; 1,483 of 2,064 bound) · curiosity genome (16,114 vs a null's 2,223 — "the one organ that beat its twin") · retina/gaze (6.4× random; unseen fonts 47.5 → 58.9 %) ·	grounding_check verified a word by its own records (the witness was itself) · the concept graph retained 3.3 % / 0 % / 0 % after a shift under every rule · LIF+R-STDP 0/5 seeds, armed lost to frozen · local corroboration 0.300 % vs 1 % (one article per claim; 30 %

	speech_v2 under a frozen judge (clears its 4-gate bar; vocabulary thin) · scripture verbatim (every negative control refused) · the exam harness (9 held-out families) · the reader service (0.29 s/book; 719 windows, 0 fallbacks) · learned index · self-census · clock + interoception · boot survival · the C++ judiciary (2,120/2,120) · growth g0-g1 (866,813 headwords) · the creature (six laws; firmness .13 → .59; reads unseen fonts; a shelf+net world closed 33/108 questions vs 12/108 with none)	fragments) · the starvation floor keyed per-candidate caused the starvation · the contradiction resolver admitted 0 pairs · 874 of the prediction ledger's rows were liveness pings · sleep never demoted a belief · exams F2/F3/F5/F6 (direction, calibration, arithmetic, code) — no faculty existed · a week of green P12 gates and a mind that could not hold a sentence

1.4 · The designs on paper today

DESIGN	WHAT IT FIXES	STATUS
FUSED-SENSES-PREREG + ADDENDUM 1 (09-17)	F3 — eyes, ears, thinking never stop; attention over thoughts only; the replay gate before threads	`written only` · 0 commits
THE THINKING ORGAN blueprint + ADDENDUM 1 THE OPEN WORLD (09-16/17); THINKING-RUNG1-PREREG	F1, F4 — sources HELD / REACHABLE / MAKEABLE; fetch and run; the five-adversary suite; a falsifiable prediction ledger	`rung 1 built, not wired` (branch unmerged)
THE CHILD'S PATH (09-16)	F7 — a rendered page → the eye → words out of a stream → grammar by compression → speech → ears → fused streams	`written only`
NEURAL-PORT-PREREG (09-16)	F6 — the growing net as a withheld organ; parity byte-exact	`was live` · at its cap · closed 09-19
P1-READER-PREREG; LANE-LAW; P0-GATE	reading whole documents; the discipline; identity	live until 09-19; the identity gate is on STATE (F9)
G0-PREREG (today)	F1, F2, F3, F4, F5, F6, F7, F9 on the smallest world with known truth	`draft for his approval`

2 · What worked, and matches the need

The need: an organism that learns words tied to pictures and grows on its own. These are the mechanisms with a measured PASS that serve it directly, and the gene or step each becomes.

MECHANISM (ERA)	EVIDENCE	IN THE GENOME
E48b's growth rule, arm A1 — the one ported: prototypes born on novelty and error under a budget;	letters retained (−1.7 / +1.7 pp); Cyrillic 0.84–0.90; byte-exact	the GROW, WIRE, SHRINK genes

delta rule; synaptic scaling; pruning under a floor; NO disuse decay. (E48b itself FAILED its conjunction on the digits — a context defect cured by D31; what PASSED was the port's parity gate.)	C++ parity over 136,416 steps; live 3 days, 6,000 units	verbatim
Homeostasis M3 (A1 → A2 → A3)	192/192 runaways contained; reconfirmed in A2; the SCALE step in E48b	SCALE gene
Context at the label (E46d, D31)	the 18-glyph conflict cured; digits .69 → .79; handwritten .19 → .66; g2 adopted	MEET: the word is the picture's context and the picture the word's
The eye tower's fixed front (E09c/d; retina/gaze in A2)	at a budget the eye's declared readings beat PCA/box/random; gaze 6.4× random; unseen fonts	RETINA gene (fixed, not learned)
Known words + statistics cut the stream (E47)	F1 0.487, 12.6× twin, 2.09× pure statistics — the one PASS on the stream road	the stream rung after G0
The cochlear front (E52)	1.000 on 553 held-out; AGC and compression load-bearing	SENSE ear, later
Curiosity genome (A2)	16,114 vs a null's 2,223 on every measurement	a WANT gene for rung 1 (what to look at next)
The jail (A2 → A3's hands); the workshop (A1)	hostile drafts refused 9/9; 442 program runs	MAKEABLE sources in rung 2+
The judge/generator split; frozen judges; the random-twin rule; pre-registration with appended amendments; the parity/oracle discipline; the P0 identity gate (A1 → A2 → A3)	never failed a control across 42 preregs; 0 mismatches over 82 files; every one of this atlas's numbers exists because of them	fixed machinery vs growing structure; twins each blind to one thing; the replay gate
gate_vitality; the frontier's exhaustion types; the self-model (A1)	detected a gate that could never pass — and nothing consumed it	ABSTAIN + the dead-end census (D39)
Retirement by sealing (A1); disposal archives before deletion (A2)	one reverted block restores an organ; nothing ever deleted	how Alpha 3 was closed today

3 · Where we are failing, and why

Ten failure classes. Each crossed at least two organisms, which is what makes it a class rather than a bug. The instance column gives the number that proves it; the cause is the mechanism; the fix names the register that carries it and how G0 tests it. **Evidence seam:** theory citations that appear in R2, R4 and the reviews were fetched on this box at the dates those files name; the

classic ones here (Popper, Goodhart, Nelson & Narens, Harnad, Grossberg, Saffran, Carey & Bartlett, Meredith & Stein) are cited from memory and were not re-verified today.

#	THE FAILURE	INSTANCES (ALPHA 1 → 2 → 3)	ROOT CAUSE	THEORY	FIX · IN G0
F1	**The closed loop over its own records**	Teacher graded vs its own lagging corpus (6/6 PENDING) → grounding_check's witness was itself; corroboration 0.3 % → 25,000 syllogisms «left me no richer»; 0 of 7,609 corroborated; «caging a bird»	no witness outside the writer: a system fed only its own output can find itself inconsistent, never wrong	Popper: knowledge is what something outside could refute; no supervision arises inside a closed loop	the open world (fetch/run), D12, an outcome he perceives · **G0:** the picture is the word's witness; truth scored outside him
F2	**The zero scored as success**	0 trades scored +0.007 vs a losing null → QD promoted «abstain, 0 positions» → abstention 75 % by design	an objective that never charges inaction	Goodhart; a conjunctive scoreboard with a participation floor; a minimal criterion before ranking (MCC)	**G0:** (i) must answer AND (vi) must abstain only on the unseen
F3	**A scheduler over the senses**	starvation measured over 1,182 passes (school 0, curiosity 0); the floor written to cure it made retry 3× more frequent for six live minutes → one duty per pass; 4× energy = 0.53× seeing; 09-18: 45 % wait, 41 % one question	perceiving was a chore that had to outrank asking	perception is continuous and predictive (Rao & Ballard; Friston); attention chooses among representations, never whether to perceive	the body prereg G8–G12 · **G0:** no scheduler exists; every moment is perceived
F4	**No consumer for «this cannot be answered»**	the frontier organ unread; senses 0/29 unlocked; a never-passable gate flagged 3 weeks → two days on `skullcap~device`, then `con~gathering` within ten minutes of the restart	a computed value nobody consumes (SOLUTIONS #163/#180)	metacognition = monitoring PLUS control (Nelson & Narens); knowing you don't know must change the next act	D39 park rule, consumed by the drive · **G0:** ABSTAIN is an answer; the null plant and Q5 test it
F5	**One statistic, no context**	F2/F3/F5/F6: no faculty to ask → E46 ×3 (18 shared glyphs read as forgetting), E43 (the wrong grain), nothing reads its own misses	P(label | shape) with nothing saying which world the shape is in	perception = expectation + evidence; the McGurk effect; context is half the likelihood	D31 (cured it) · **G0:** MEET makes each side the other's context
F6	**Forgetting by design, then**	plasticity 0/63; pruning 0/32 vs random; the ceiling was the feature	shrinking by DISUSE is forgetting; a	stability–plasticity (Grossberg; ART → GWR); loss of	prune the wrong and redundant; budget, never

	growth that stops	vector → the concept graph 3.3 % / 0 % / 0 % → E48's cap ate the old; decay ×0.064; the live net at N_max, 1 % pruned	cap is not a cycle; rules cannot learn what features do not carry	plasticity as a second failure (Dohare); NEAT: ablate growth, failure 0 → 80 %	cap; a fixed retina · **G0:** those genes; marks (iv) (v); D41
F7	**The pre-cut word — a symbol with no body**	1.27 M senses, own_perception 1 word of 20; «a dictionary reports what is distinctive, not what is true» → 27,664 words as counts; the eye and the reader never met; R1's gloss road overturned by his 09-15 study	language as symbol lookup; a word never beside the thing it names	symbol grounding (Harnad); statistical segmentation with the referent present (Saffran; fast mapping; mutual exclusivity); multisensory binding (Meredith & Stein)	the present · **G0 is this:** a picture and its word in one moment; later the stream and order
F8	**The instrument lied first**	FDR 83–89 % from noise; cluster-SE 72.8 % false positives → E05's port bugs; E29's random θ; E44's plant shared with every negative; a twin blind to everything (#228)	measuring the thing before the instrument	pre-registration, negative controls, null twins, ceilings before bars — the 230-lesson corpus	carry it whole · **G0:** the toy first, gate G-a, twins, the plant with its own budget, the replay gate
F9	**Built before it could learn**	«a week of green gates, a mind that could not hold a sentence» → P0 froze the life byte-for-byte; every learner lived at the edge under a withheld flag; «going to c++ with not working system is costing us»	an identity gate on STATE, not on RULES; ports before proof	a genome IS that separation: fixed machinery, growing structure; identity by replay, not by frozen state	Python first; C++ for what passed · **G0:** mark (vii)
F10	**Self-modification that recombines, never invents**	3 of 33 knobs; the LLM mutator never beat random → E29's evolved encoding lost to 48 random draws; E49 bought nothing; «self-modification recombines an existing alphabet»	search over a fixed alphabet on a substrate that cannot learn; evolution before development	open-ended = novel AND learnable (R4); fixed-code systems cannot be open-ended (Banzhaf); NEAT grows with speciation, starting minimal; MCC	development first, a lineage only on a learning substrate; E29's control forever · **G0:** D-G1, mark (ix)

> Two more facts belong beside the table. The organisms' method churned: Alpha 1 re-pivoted eleven times in thirteen days; Alpha 2's organ-by-organ body (200 organs, 737 commits in ten days) was retired after a week; Alpha 3 ran 56 experiment directories in seven days. And the box changed under the plan: on 09-12 it had no numpy, which is why R4 specified its ALife tests in C++ and why the organism and its organs were kept in C++ under the P0 identity gate; most experiment

> directories are Python, and numpy reached the box within days — today Python 3.14 with numpy, PIL and scipy is what makes a Python genome possible now.

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4 · The theoretical solution

An organism is a **genome** (rules, not structure) run by fixed **machinery**, living in a **world that answers back** (a present where a word meets its picture; questions scored outside it), with a **body that always perceives** (no scheduler over senses), a **mind that can say «I do not know»** (a consumer for every dead end), **growth that never eats** (a budget; prune the wrong and the redundant), and a **discipline that measures its own instrument first**. Each of the ten failures is the absence of one clause. G0 tests the first seven on the smallest world where the truth is known; the body's threads, the open world and the lineage are rungs on the same substrate, in that order, each behind its own gate.

What would change this: a G0 organism that passes (i) and (ii) but fails (vi) would say the substrate learns and the metacognition gene is wrong, not the design; a G0 that fails (ii) — the shuffled twin reading above chance — would say the association is geometry, not co-occurrence, and the whole «present» clause is false for this world; a random genome in the top quartile (ix) would say E29 again, and the genome's numbers are decoration.

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5 · The whole working mechanism, as one loop

This is his research of 2026-09-19 (eight sections) drawn as one machine. The loop runs left to right and back; the genome and the grow/prune rules sit under everything because every box above is meant to be *grown*, not written. The coloured band names what exists in Alpha 3 today, so the gaps are visible in the same picture.

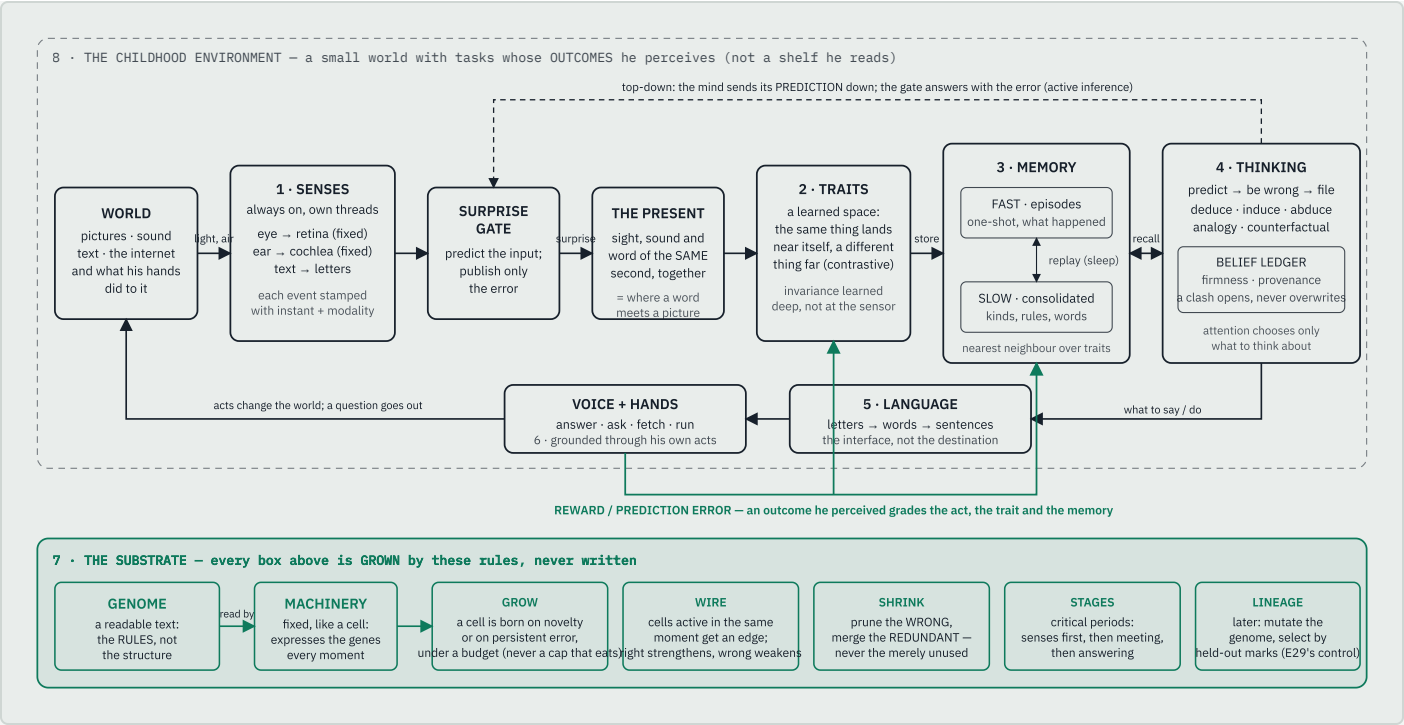


Figure A. The mechanism his research describes, as one loop. Numbers 1–8 are his research's sections. Green is the substrate the genome experiment builds first; the loop above it is what the substrate must be able to grow, rung by rung. The dashed top arrow is the one thing Alpha 3 never had: a prediction sent DOWN to an organ.

BOX IN FIGURE A	ALPHA 3 TODAY (CLOSED 09-19 14:44)	RUNG 0 OF THE GENOME	RUNG 1–2	LATER GENE FAMILY
senses always on	one thread, one duty per pass; the eye every 50 passes; ear not live	one stream, single process (deterministic)	rung 1: his pictures + words	SENSE ear (E52's cochlear front); threads under the replay gate
surprise gate	none	BIRTH novelty is a gate on the LEARNER only	—	PREDICT gene: publish only the error
the present	never: eye and reader never met	**yes** — a picture and its word in one moment	rung 1: real pictures	sound in the same moment
traits	a sealed tower + prototypes keyed to exam labels	prototypes born from the retina's features	—	a contrastive gene (same thing near, different far)
memory fast / slow + replay	counts and verbatim ledgers; no replay	cells + edges (slow only)	—	EPISODE gene + SLEEP gene (replay)
thinking + belief ledger	one syllogism, «left me no richer» ×25,000; predictions 0	RECALL + ABSTAIN («I do not know»)	—	PREDICT, CLASH, ANALOGY genes; the open world

language	letters and word counts; no sentence	a word = letters (the ear for text)	—	ORDER gene (sentences)
voice + hands, reward	frames; hands run programs; no outcome grades anything	answers scored OUTSIDE him	rung 1: his questions	REWARD gene: an outcome he perceives
grow / prune	at its cap (6,000 units), 1 % pruned	**yes** — the whole point	rung 2: lineage	MERGE + the grow/prune cycle
childhood environment	a shelf he reads; exams sat by the seat	a world with known truth	rung 1: his world	tasks with consequences

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6 · The plan: the genome experiment, rung 0

How the experiment is performed. The full pre-registration with every bar is `docs/G0-PREREG.md`; this is the same thing drawn.

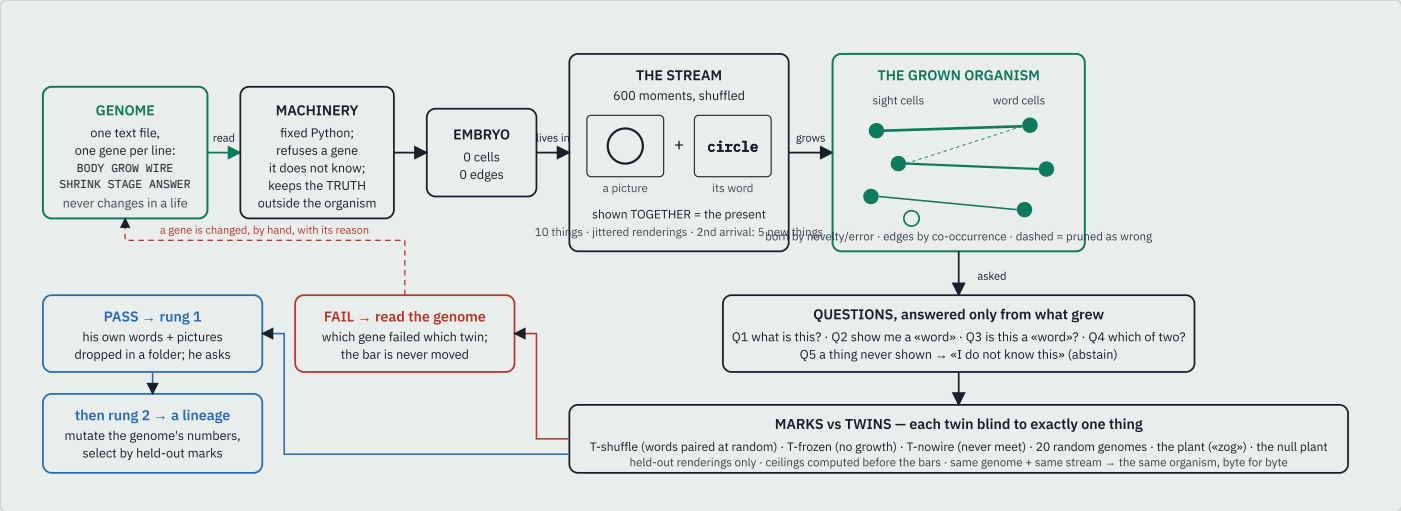


Figure B. How the experiment is performed. The organism is never written: the genome is read by fixed machinery into an embryo, and the stream grows it. The questions are answered from the grown cells and edges alone; the truth used to score them stays outside the organism. Twins each lack one thing so a pass names what carried it.

The genome, as he will read it

```
SEED   20260919
# BODY — what senses exist and what areas the organism has
SENSE  eye    24x24 grey        # a picture is 576 numbers
RETINA centre pool3 unit        # FIXED front, not learned: centre on the ink, 3x3 pool ->
SENSE  ear    letters 8         # a word is up to 8 letters, one-hot per position, unit len
AREA   sight  from retina
AREA   words  from ear
MEET   sight  words             # rung 0: the meeting is EDGES ONLY (sight cells x word cel
```

```
# GROW — when a cell is born (E48b's proven rule)
BIRTH  novelty  cos<THETA_N     # no cell resonates -> a new cell holds the input; THETA_N r
BIRTH  error    misfires 3      # a cell keeps answering wrongly -> a new cell beside it
BUDGET 400 per arrival          # never replace a cell; refusals are printed
NMAX   2000
# WIRE — how cells connect
KWTA   5                        # five winners per area per moment
HEBB   window 1                 # cells active in the SAME moment across areas get an edge
DELTA  lr 0.05                  # the edge grows when the pairing is right, shrinks when wr
# SHRINK — the wrong and the redundant; never the merely unused
PRUNE  w<0.02 after 3 chances
MERGE  cos>THETA_M same-answer
SCALE  every 1000 moments       # synaptic scaling (homeostasis)
# STAGE — development in order
STAGE  1 senses   moments 0..200
STAGE  2 meet     moments 200..
# ANSWER — how a question is answered from what grew
RECALL k=1                      # activate one side; read the strongest edge to the other
ABSTAIN second/best > 0.85      # too close to call -> "I do not know this"
```

«It works» means, in one sentence: grown from the genome alone, the organism names held-out pictures at 80 % or better and shows the right picture for a word, reads at chance when the words were shuffled during its development, keeps the first ten things after five new ones arrive, and says «I do not know this» for a thing it was never shown.

D-G1 — development only in rung 0; evolution (a lineage) in rung 2. Recommended: yes. E29 showed that evolving without a working substrate rewards luck.

D-G2 — rung 0's world is synthetic with known truth; his own pictures and words come in rung 1 through a folder door. Recommended: yes. The toy must be solvable before the world is real.

Rung 0 is Python 3.14 + numpy, one process, one afternoon: the machinery and the renderer, the toy gate (a 2-thing world must reach 1.0), gate G-a (the novelty threshold measured from the world's own cosine bands before it is declared), then the 10-thing life, the twins, the plant, and a VERDICT with every number. Nothing from this page is built until his yes.